❶ Randomly generate the initial training data.

❷ Record incumbent value and retrain model.

❸ Find the point maximizing the acquisition score, query the objective function, and update the training data.

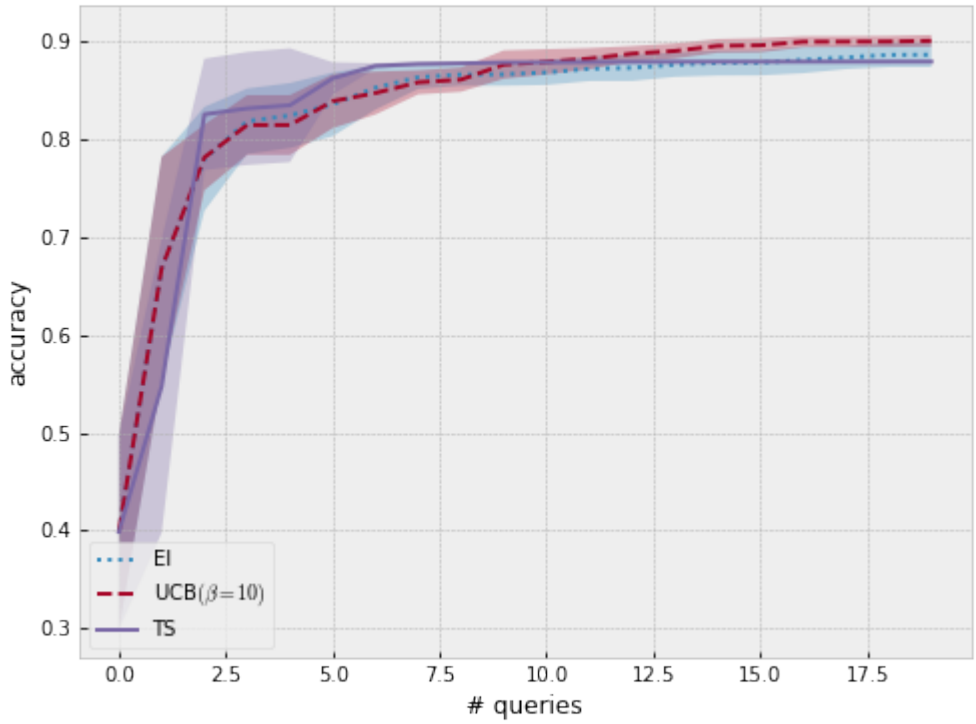


Figure A.13 Progress made by Thompson sampling. The policy is comparable to Expected Improvemen and slightly worse than the best version of Upper Confidence Bound.

Figure A.13 shows the optimization performance of Thompson sampling. We see that the policy makes significant progress at the beginning and is comparable to Expected Improvement from Chapter 6 and slightly worse than the best version of UCB.

A.6 Chapter 6: Leveraging information theory with entropy-based policies

There are two exercises in this chapter:

1. The first is mathematical in nature, covering a variant of binary search in which prior information could be taken into account when making decisions.
2. The second walks us through the process of implementing Max-value Entropy Search in the hyperparameter tuning problem seen in previous chapters.

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